ANIMAL_PERFORMANCE.xlsx

This file contains the animal performance measured during the trial in each experimental pen.

Column	Name of the column	Description	Unit
A	PEN	Pen number	
B	TREATMENT	Experimental treatment	
C	BW_D1	Average body weight of birds at the beginning of Starter phase	g
D	BW_D10	Average body weight of birds at the end of Starter phase	g
E	BW_22	Average body weight of birds at the end of Grower phase	g
F	BW_35	Average body weight of birds at the end of Finisher phase (= end of the trial)	g
G	ADG_D1_D10	Average daily body weight gain during the Starter phase	g/day
H	ADG_D11_D22	Average daily body weight gain during the Grower phase	g/day
I	ADG_D23_D35	Average daily body weight gain during the Finisher phase	g/day
J	ADG_D11_D35	Average daily body weight gain during the Grower+Finisher phases	g/day
K	ADG_D1_D35	Average daily body weight gain during the complete trial	g/day
L	ADFI_D1_D10	Average daily feed intake during the Starter phase	g/day
M	ADFI_D11_D22	Average daily feed intake during the Grower phase	g/day
N	ADFI_D23_D35	Average daily feed intake during the Finisher phase	g/day
O	ADFI_D11_D35	Average daily feed intake during the Grower+Finisher phases	g/day
P	ADFI_D1_D35	Average daily feed intake during the complete trial	g/day
Q	FCR_D1_D10	Feed conversion ratio during the Starter phase	kg feed/kg body weight gain[1]
R	FCR_D11_D22	Feed conversion ratio during the Grower phase	kg feed/kg body weight gain[1]
S	FCR_D23_D35	Feed conversion ratio during the Finisher phase	kg feed/kg body weight gain[1]
T	FCR_D11_D35	Feed conversion ratio during the Grower+Finisher phases	kg feed/kg body weight gain[1]
U	FCR_D1_D35	Feed conversion ratio during the complete trial	kg feed/kg body weight gain[1]
V	MORTA_D1_D35	Mortality rate during the complete trial	%

[1] including dead birds

BODY_COMPOSITION.xlsx

This file contains the data on body composition measured by dual-energy X-ray absorptiometry (DXA-scan) on one bird per pen.

Column	Name of the column	Description	Unit
A	PEN	Pen number	
B	TREATMENT	Experimental treatment	
C	BW_D22	Body weight of the bird at the end of the Grower phase	g
D	BW_D35	Body weight of the bird at the end of the Finisher phase (= end of the trial)	g
E	WATER_D22	Water content of the bird at the end of the Grower phase	%
F	PROTEIN_D22	Protein content of the bird at the end of the Grower phase	%
G	ASH_D22	Ash content of the bird at the end of the Grower phase	%
H	LIPID_D22	Lipid content of the bird at the end of the Grower phase	%
I	WATER_D35	Water content of the bird at the end of the Finisher phase (= end of the trial)	%
J	PROTEIN_D35	Protein content of the bird at the end of the Finisher phase (= end of the trial)	%
K	ASH_D35	Ash content of the bird at the end of the Finisher phase (= end of the trial)	%
L	LIPID_D35	Lipid content of the bird at the end of the Finisher phase (= end of the trial)	%

SCFA.xlsx

This file contains the data on caecal short chain fatty acids measured at the end of the trial (D35) on one bird per pen.

Column	Name of the column	Description	Unit
A	PEN	Pen number	
B	TREATMENT	Experimental treatment	
C	TOTAL_SCFA	Short-chain fatty acid content in the caecal content	mM/g of caecal content
D	ACETATE_RATIO	Proportion of acetate in total short chain fatty acids	%
E	PROPIONATE_RATIO	Proportion of propionate in total short chain fatty acids	%
F	BUTYRATE_RATIO	Proportion of butyrate in total short chain fatty acids	%
G	VALERATE_RATIO	Proportion of valerate in total short chain fatty acids	%
H	CAPROATE_RATIO	Proportion of caproate in total short chain fatty acids	%
I	ISO-BUTYRATE_RATIO	Proportion of iso-butyrate in total short chain fatty acids	%
J	ISO-VALERATE_RATIO	Proportion of iso-valerate in total short chain fatty acids	%

LITTER_COMPOSITION.xlsx

This file contains the data on litter weight and composition at the end of the trial (D35) in each experimental pen.

Column	Name of the column	Description	Unit
A	PEN	Pen number	
B	TREATMENT	Experimental treatment	
C	LITTER_WEIGHT	Total weight of the litter in the pen at the end of the trial	kg/pen
D	LITTER_ACC	Litter accumulation during the trial[1]	g/kg body weight gain
E	pH	pH of the litter	
F	DM	Dry matter content of the litter	%
G	N	Total nitrogen content of the litter	g/kg fresh matter
H	TAN	Total ammoniacal nitrogen content of the litter	g/kg fresh matter
I	P	Total phosphorus content of the litter	g/kg fresh matter
J	K	Total potassium content of the litter	g/kg fresh matter
K	TAN_N_RATIO	Ratio between ammoniacal and total nitrogen contents	g/g
L	N_P_RATIO	Ratio between total nitrogen and total phosphorus contents	g/g
M	N_K_RATIO	Ratio between total nitrogen and total potassium contents	g/g

[1] Calculated as: (Final litter weight – initial weight of wood shavings) / total body weight gain of the pen including dead birds.

MASS_BALANCE.xlsx

This file contains the data on nitrogen (N) mass balance performed in each pen on the complete experimental period (D1-D35).

Column	Name of the column	Description	Unit
A	PEN	Pen number	
B	TREATMENT	Experimental treatment	
C	TOTAL_N_INPUTS	Total N amount entering the pen[1]	kg N per pen
D	TOTAL_N_INTAKE	Total N intake from feed	kg N per pen
E	TOTAL_N_CHICKS	Total N amount in chicks at D1	kg N per pen
F	TOTAL_N_BEDDING	Total N amount in wood shavings at D1	kg N per pen
G	TOTAL_N_EXCRETION	Total N excretion in the pen during the trial	kg N per pen
H	TOTAL_N_MANURE	Total N amount in the litter at the end of the trial	kg N per pen
I	TOTAL_N_BROILERS	Total N in broilers[2]	kg N per pen
J	TOTAL_N_VOLATILIZATION	Total N volatilization for the pen[3]	kg N per pen
K	TOTAL_N_INPUTS_kgBW	Total N amount entering the pen[1]	g N per kg body weight[2]
L	TOTAL_N_INTAKE_kgBW	Total N intake from feed	g N per kg body weight[2]
M	TOTAL_N_CHICKS_kgBW	Total N amount in chicks at D1	g N per kg body weight[2]
N	TOTAL_N_BEDDING_kgBW	Total N amount in wood shavings at D1	g N per kg body weight[2]
O	TOTAL_N_EXCRETION_kgBW	Total N excretion in the pen during the trial	g N per kg body weight[2]
P	TOTAL_N_MANURE_kgBW	Total N amount in the litter at the end of the trial	g N per kg body weight[2]
Q	TOTAL_N_BROILERS_kgBW	Total N in broilers[2]	g N per kg body weight[2]
R	TOTAL_N_VOLATILIZATION_kgBW	Total N volatilization for the pen[3]	g N per kg body weight[2]

[1] chick + feed + wood shavings.

[2] including dead birds.

[3] calculated as: C – (H + I)